

NEXUS GUILDS

AI Foundational

From Professional to AI Engineer

Week 2: The Landscape — AI Models & Selection
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Pre-Reading: ~1.5 Hours | In-Class Session: 2 Hours

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Andrés's New Challenge



The Three-Tool Problem

Marketing loves ChatGPT's creativity. Engineering trusts Claude for code reviews. Legal prefers Gemini's citations and its massive context window for contract analysis.

The CFO walks in and asks: "Why do we need three AI subscriptions? Can't we just pick one?"

Andrés knows the answer is not that simple. He needs a framework — a clear, defensible way to recommend the right model for the right task. His boss wants an AI Tool Selection Matrix by Friday.

This week, you build that matrix alongside Andrés.



Bring your Prompt Playbook from Week 1. Prepare at least 3 CRFT prompts: one creative, one code, one analytical.

Week 2 Overview



Part 1: Pre-Reading (~1.5h)

- Why No Single Model Wins
- The Six Model Families
- 6-Dimension Selection Framework
- Model Tiers & Pricing



Part 2: In-Class Session (2h)

- The Three-Tool Problem opening
- Model families & framework review
- Exercise 2.A: Model Comparison
- Exercise 2.B: Build your Selection Matrix
- Peer validation & debrief

75% Practical

PART 1

Pre-Reading: The AI Model Landscape

Complete before your in-class session | Estimated time: ~1.5 hours

Why No Single Model Wins

Reading 1 | ~20 minutes



The Restaurant Analogy

You wouldn't ask "What is the best restaurant?" without context. Best for a romantic dinner? A quick lunch? A group of 20? On a budget?

AI model selection works the same way. The "best" model depends on what you are optimising for: quality, speed, cost, context size, specialisation, or ecosystem.

The right question is never "Which model is best?" — it is always "Which model is best for this specific task, with these constraints?"

The six model families you need to know:

Claude

GPT

Gemini

DeepSeek

Llama

Grok

NO

The Six Model Families

Reading 2 | ~40 minutes



Claude

Anthropic

Code, analysis, 200K context,
extended thinking



GPT

OpenAI

Creative writing, multimodal,
ecosystem breadth, plugins



Gemini

Google

1M+ context, web grounding,
multimodal (text/image/video)



DeepSeek

Chinese AI Lab

Cost-effective, open-weight, strong
code & maths



Llama

Meta

Open-source, fine-tuning,
on-premise, community



Grok

xAI

Real-time data, reasoning, less
filtered responses

Each family has distinct strengths. No single model dominates all categories.

Claude (Anthropic)

Opus 4.5 | Sonnet 4 | Haiku 3.5

Where Claude Excels

- Long document analysis (200K+ token context)
- Code generation and review
- Structured analytical tasks
- Extended thinking for complex reasoning

The Tiering System

Opus — Most capable, highest cost

Sonnet — Workhorse (most tasks)

Haiku — Fastest, cheapest



When to Reach for Claude

150-page contract to analyse. Code review of a multi-file project. Step-by-step complex business decisions. Structured, precise outputs with minimal hallucination.

GPT (OpenAI)

GPT-4o | o3 | o4-mini

Where GPT Excels

- Creative writing and brainstorming
- Multimodal tasks (images + text)
- Function calling (invoke external tools)
- Broadest plugin/integration ecosystem

Reasoning Models

o3 and o4-mini "think through" problems before responding. Strong at maths, science, and coding. Trade-off: slower due to extra internal computation.



When to Reach for GPT

Creative marketing copy. Process images + text together. Connect to third-party tools via function calling. Team already uses ChatGPT's familiar interface.

Gemini (Google)

Gemini 2.5 Pro | Gemini 2.5 Flash

Where Gemini Excels

- 1M+ token context (5–7 full books!)
- Native web grounding with cited sources
- Multimodal: text, images, audio, video
- Deep Research with verified citations

Flash vs. Pro

Flash — Speed & efficiency (like Haiku).
Best for high-throughput.

Pro — Full power. The right choice for
most professional tasks.



When to Reach for Gemini

500–page document that exceeds other models' limits. Research report with verifiable sources. Mix of text, images, and video in one query. Legal team wants grounded claims.

DeepSeek, Llama & Grok



V3 | R1 (reasoning)

Cost-effective with premium performance. Open-weight (self-host for privacy). Strong code & maths benchmarks.



Self-hosting needs GPU infrastructure. "Free weights" ≠ free to run.



Llama 4 Scout | Maverick

Open-source with massive community. On-premise for data residency. Fine-tune on proprietary data. Avoid vendor lock-in.



Requires technical team for deployment and maintenance.



Grok (xAI)

Grok 3 | Grok 3 Mini

Real-time data access & trends. Strong reasoning. Less filtered responses for legitimate business needs.



Less filtered = stronger output validation needed. Check content policies.

The 6-Dimension Framework

Reading 3 | ~30 minutes



Quality

Accuracy, reasoning, creativity, coherence



Speed

Time to first token, total generation time



Cost

Price per million tokens at expected volume



Context Window

Max tokens per request (document length)



Specialisation

Strength in code, creative, analysis, maths



Ecosystem

APIs, docs, integrations, community, tools

No model wins on all six — that is the fundamental insight that makes this framework useful.

How to Use the Framework

1

Define Requirements

List what matters most for this task. Rank the six dimensions by importance for your use case.

2

Evaluate Candidates

Test 2-3 models against your actual task. Use your own prompts, data, and evaluation criteria.

3

Document & Recommend

Record findings in the AI Tool Selection Matrix. Include primary and fallback models for each task.



The Weighted Decision

Not all dimensions are equally important for every task. A creative writing task might weight Quality (40%) and Speed (20%) heavily but care little about Context Window (5%). A legal document analysis might weight Context Window (35%) and Quality (35%) but accept higher cost. Define your weights before you evaluate — it prevents post-hoc rationalisation.

Quick-Reference Comparison

Model	Quality	Speed	Cost	Context	Specialisation	Ecosystem
Claude	★★★★★	★★★	★★★	200K+	Code, Analysis	★★★★
GPT	★★★★★	★★★★	★★★	128K	Creative, Multi	★★★★★
Gemini	★★★★	★★★★	★★★★	1M+	Research, Multi	★★★★
DeepSeek	★★★★	★★★	★★★★★	128K	Code, Maths	★★★
Llama	★★★★	★★★	★★★★★*	Varies	Fine-tunes	★★★★
Grok	★★★★	★★★★	★★★	128K	Real-time, Reason	★★★

**Llama weights are free, but self-hosting requires infrastructure investment.*

This is a snapshot from early 2026. The framework (how to evaluate) is durable. The specific ratings are perishable.

Model Tiers & Pricing

Reading 4 | ~20 minutes



Flagship

Highest capability. Complex analysis, high-stakes decisions.

Opus 4.5, GPT-4o, Gemini 2.5 Pro, Grok 3



Balanced

Strong capability at moderate cost. The workhorse for most professional work.

Sonnet 4, o3, DeepSeek-V3



Lightweight

Fastest, cheapest. Classification, summarisation, high-volume tasks.

Haiku 3.5, o4-mini, Flash, Grok 3 Mini



The 80/20 Rule of Model Tiers

80% of AI tasks can be handled by the balanced tier. This tiering strategy alone can reduce AI costs by 40-60% compared to routing everything through the flagship.

The Cost Iceberg

Token pricing is only the visible part of the cost.

API Costs

Per million tokens (input & output priced separately). Output tokens typically cost 3-5x more than input.

Infrastructure Costs

For self-hosted models: GPU servers at \$10K-50K+ per year each.

Integration Costs

Developer time to build and maintain API integrations, error handling, monitoring.

Switching Costs

Once prompts, workflows, and integrations are built for one model, switching requires rework. This is vendor lock-in.

Key takeaway: Choosing the right tier within a family is just as important as choosing the right family. NG

PART 2

In-Class: The Landscape

Model Comparison & Selection Matrix | 2 Hours | 75% Practical

Session Plan

0:00–0:10

Opening: "The Three-Tool Problem" + Playbook highlights

Conceptual

0:10–0:25

Core Teaching: Model families, 6-Dimension Framework

Conceptual

0:25–1:10

Exercise 2.A: Model Comparison Challenge (3 prompts × 3 models)

Practical

1:10–1:50

Exercise 2.B: Build & peer-validate your Selection Matrix

Practical

1:50–2:00

Debrief: Did the "best" model vary by task?

Practical

By the end of tonight, Andrés has his answer. So will you.

Exercise 2.A: Model Comparison

45 minutes | Work with a partner | 3 prompts × 3 models

1

Creative Task

200-word product launch for "FitMind" AI fitness app. Energetic, professional tone.

2

Code Task

Python function: top 3 scorers from list of dicts. Handle edge cases + pytest tests.

3

Analysis Task

Retail: online -15%, in-store +8%, CSAT steady 4.2/5. Top 3 explanations + data sources.

Run each prompt on: Claude | ChatGPT | Gemini (minimum)

Rate each output 1-5:

Quality

Speed

Format

Usefulness

Surprise Factor

Exercise 2.B: Selection Matrix

40 minutes | Build + Peer Validate

Task Category	Priority Dims	Recommended	Fallback	Justification
Python code review (security focus)	Quality #1 Specialisation #2	Claude Sonnet 4	GPT-4o	Claude's code analysis + 200K context for multi-file reviews
Marketing copy (product launch)	Quality #1 Speed #2	GPT-4o	Claude Sonnet 4	GPT produces more engaging, natural-sounding creative text
Contract analysis (200+ pages)	Context #1 Quality #2	Gemini 2.5 Pro	Claude Opus 4.5	Gemini's 1M+ context handles full contracts in one pass

Build your own matrix with at least 5 task categories from your work, then swap with a partner.

Peer Validation Scenarios:

1. Legal team: 200-page contract, overnight
2. Startup: 50K support tickets/month
3. CEO: compelling board vision document
4. Data team: fine-tune on proprietary data

Week 2 Deliverable



AI Tool Selection Matrix

A personalised decision framework mapping task categories to recommended AI models, with priority dimensions, fallback models, and justifications. Peer-tested with at least 3 scenarios.

Created during: In-class exercises (throughout)

Your Growing Portfolio



Week 1

AI Engineer's Compass



Week 1+

Prompt Playbook
(ongoing)



Week 2

AI Tool Selection Matrix

Looking Ahead: Week 3



Andrés presents his Matrix to the CFO. Response: "Exactly what we needed. But here's the next challenge — our engineering team says building AI integrations takes too long. They heard about tools that let you describe an app in plain English and it builds itself. Is that real?"

It is real. Next week: the full spectrum of AI-assisted development. From no-code builders (Replit Agent, Bolt.new) to agentic coding environments (Cursor, Claude Code). You build something real.

Week 3: AI-Assisted Development

- No-code app builders: describe what you want, watch it appear
- Agentic coding: AI as your pair programmer
- Build a real, working prototype — with AI doing most of the coding

Setup before Week 3: Create Replit account | Install Cursor | GitHub account ready

***"The expert in anything
was once a beginner."***

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Source Meridian